Type and Complexity Signals in Multilingual Question Representations: A Diagnostic Study with Selective Control Tasks

Robin Kokot* ROBIN.KOKOT@STUDENT.KULEUVEN.BE

Faculty of Engineering Science, KU Leuven, Leuven, Belgium

Abstract

We investigate how Glot500-m, a multilingual transformer model, encodes structural linguistic properties of questions. Our experiments use binary classification and continuous regression tasks designed to probe the encoding of question types and morphosyntactic complexity in seven languages. We compare the performance of probes to statistical term frequency baselines and the fine-tuned model, using selectivity controls with shuffled labels to contextualize our results. We show that questions in languages with more explicit type markers can be classified accurately using subword frequency features, while those with more implicit strategies benefit from contextual embeddings. We introduce regression probing for complexity metrics and demonstrate that traditional ML algorithms trained on TF-IDF features perform well on composite complexity scores. However, neural probes often achieve better selectivity than linear regression or gradient boosting. Individual submetrics show noticeable patterns: statistical models outperform probes on structural metrics like tree depth and subordinate chain length, while neural probes perform better at predicting token count, with relatively flat performance curves across layers indicating limited layer-wise specialization. We also note that probes on frozen encoder representations often outperform the fine-tuned model on all regression tasks, suggesting that end-to-end updating may degrade performance on linguistic regression labels.

1. Introduction

Recent work on massively multilingual LLMs shows significant progress in pre-trained models being able to handle standard downstream tasks like translation, question answering, and syntactic analysis on anywhere from dozens to hundreds of languages (Liu et al. 2020, Clark et al. 2020, Conneau et al. 2020, Doddapaneni et al. 2025). Interest in the abilities of these models extends beyond traditional benchmarks to more nuanced questions about how these models process multilingual phenomena like code-switching and lexical borrowing, or engage in linguistic reasoning across languages (Muller et al. 2023, Beguš et al. 2025). Understanding how these models represent linguistic features is critical for improving their reliability and performance in multilingual settings. We investigate this issue through a series of experiments with diagnostic classifiers designed to reveal whether pre-trained contextual encoders adequately represent linguistic features related to the meaning of interrogative sentences and their morphosyntactic complexity.

First, we ask whether Glot500-m (Imani et al. 2023), a pre-trained multilingual encoder model, supports better generalization on sentence-level tasks than traditional statistical representations. We develop a set of baseline experiments comparing diagnostic classifiers trained on frozen LLM embeddings to classical ML algorithms trained on subword TF-IDF features. Next, we investigate how linguistic signal strength varies across LLM layers, identifying whether properties such as sentence type, number of tokens, dependency tree depth, and lexical density are more salient at early, intermediate or final layers of the encoder. Finally, we connect these two questions to a broader methodological issue of whether diagnostic classifiers learn true linguistic composition or memorize patterns in the data.

To address the first two questions, we develop a methodological framework inspired by linguistic probing methods (Conneau et al. 2018, Ravishankar et al. 2019a, Tenney et al. 2019b). This includes a set of custom tasks designed to compare performance of models trained on different text features and isolate signals encoded in different layers of the model. We address the issue of meaningful generalization by comparing performance on genuine labels against three different control sets with shuffled labels, allowing us to tell apart learning from memorization, and to quantify selectivity of each studied configuration (Hewitt and Liang 2019, Pimentel et al. 2020).

We operationalize these questions by introducing two linguistically motivated tasks that require processing complete sentences. The first is question type classification, where the goal is to distinguish between polar (yes/no) and content (wh-word) questions. This simple binary distinction is fundamental across most of the world's languages, but it is often realized through different grammatical mechanisms, from interrogative particles and punctuation to word order changes, making it an appropriate test case for multilingual representation capabilities.

The second task is linguistic complexity metric regression, which requires predicting a continuous normalized score derived from linguistic features that contribute to the functional load of a sentence, or so-called processing difficulty (Hawkins 2007, Sinnemäki 2011). These features include structural metrics (average dependency length, maximum tree depth, subordinate clause count), lexical features (lexical density, token count), and verbal complexity measures. Additionally, we vary task granularity by comparing the performance of models trained on individual submetrics to models trained on an unweighted sum of all metrics as a working model of abstract linguistic complexity. By examining both the composite score and individual submetrics, we can point out which aspects of linguistic complexity are most effectively encoded in model representations.

In the following section, we review previous work on the evaluation of text representations with diagnostic classifiers, focusing on multilingual settings. In section 3, we describe our approach to question type classification and linguistic complexity prediction, summarize the motivation for our tasks and metrics, and introduce the proprietary dataset.[1] Section 4 presents our experimental framework.[2] We report the results in section 5, and provide further analysis in section 6. We conclude with the theoretical implications of our work and interesting directions for future research.

2. Diagnostic Classifiers for Sentence Representations

Contextualized representations of natural language text are studied using various approaches, from visual methods such as saliency mapping and attention visualization (Strobelt et al. 2018, Liu et al. 2019b) to quantitative performance-based evaluations. Although visual approaches gained considerable support in early work on LLM interpretability, the efforts progressively shifted towards quantifiable analysis with interpretable metrics. Moreover, the lack of gold standard references for interpretation of representations generated by neural models led to the development of diverse tests for the presence of targeted properties and structures.

Linguistic probing tasks offer a systematic approach to understanding representational capabilities of language models by training simple classifiers to predict discrete properties from embedding spaces (Adi et al. 2017). Early work by Köhn (2015) and Shi et al. (2016) demonstrated the ability of 'probes' to extract syntactic information related to POS tags, word content, voice, tense, and canonical order from both contextualized embeddings (LSTMs) and static word representations (GloVe, Word2Vec).

Following these initial successes, probing methods have been used to evaluate suites of downstream NLP tasks using frameworks like SentEval (Conneau and Kiela 2018) and LINSPECTOR

1. https://github.com/rokokot/qtype-eval.git
2. https://huggingface.co/datasets/rokokot/question-type-and-complexity

(Şahin et al. 2020). These efforts positioned probing as a complementary method to visualization and enabled more targeted assessment of model capabilities. Ravishankar et al. (2019b) describe early challenges in probing models pre-trained on multilingual corpora, and report results on tasks such as dependency tree depth prediction, bigram shift detection, and subject-object identification. For one, controlling for different probing dataset sizes and quality of annotation efforts is difficult and often impossible when working with low resource languages. Similar concerns also apply to pre-training due to small datasets, script differences, and varied tokenization needs.

Prior work on probing often highlights certain patterns in how linguistic information is distributed across network layers. Tenney et al. (2019a) suggest a monotonic progression from syntactic (simple) processing in lower layers to semantic (more complex) understanding in higher layers. Liu et al. (2019a) conduct similar experiments on different tasks but cautiously report that higher levels of the encoder are more task specific with respect to pre-training objectives and lower more transferable between probing tasks. This hierarchical organization has become something of conventional wisdom for transformer encoder models, with lower layers of monolingual encoders often outperforming higher ones on tasks like part-of-speech tagging, while higher layers excel at tasks like word-sense disambiguation and tense prediction (Jawahar et al. 2019a, Rogers et al. 2020). However, it is an open question whether multilingual contexts warrant this kind of organization, especially when the boundary between syntactic and semantic information becomes blurred due to different grammatical mechanisms.

Another set of questions concerns the differences in reporting probing results. As approaches became more widespread, significant discrepancies emerged in how task performance was documented and interpreted across studies. Serikov et al. (2022) illustrate these challenges by comparing studies where different accuracy scores on the same tasks led to similar characterizations of model capabilities (Ravishankar et al. 2019b, Tikhonova et al. 2023). Multilingual contexts make these issues more pronounced, including additional challenges related to typological diversity and the need for language-agnostic evaluation approaches. Universal Dependencies (UD) Treebanks (Nivre et al. 2020) are a framework for consistent grammatical annotation across 150 languages that recently found new applications in probing research. Hewitt and Manning (2019) and Serikov et al. (2022) describe how standardized syntactic and morphological annotations can be used to isolate individual linguistic phenomena and design scalable probing experiments.

What is often missing from probing studies examining intermediate neural network representations is the broader context of evaluation. In particular, we are missing comparisons with non-neural baseline representations. This gap makes it difficult to determine whether observed properties are unique to neural architectures or merely reflect patterns easily captured by simpler models. Our work addresses this gap by comparing probing results with traditional ML baselines using frequency-based sentence representations (TF-IDF), providing a more comprehensive perspective on representational learning. By establishing these baselines, we can determine whether probes recover linguistic structure beyond surface-level statistics. Hewitt and Liang (2019) and Belinkov (2022) survey and address these concerns by advocating for the use of control tasks that associate target labels with random outputs. They argue that good diagnostic classifiers achieve high accuracy on genuine tasks and low accuracy on controls, which leads to more reliable evaluation methods and a broader range of probing objectives.

These extensions have recently been implemented in projects that emphasize practical applications of probing methods. For example, Zhu et al. (2022) show that the ability of transformer models to strongly encode linguistic property signals correlates with their fine-tuning potential across linguistic tasks. Similarly, Waldis et al. (2025) established connections between structural probing and safety alignment, finding that models with stronger encoding of specific linguistic features demonstrate more predictable and controllable behavior. These developments suggest a shift from viewing

probes merely as analytical tools to seeing them as guides for model selection, ability prediction, and alignment verification.

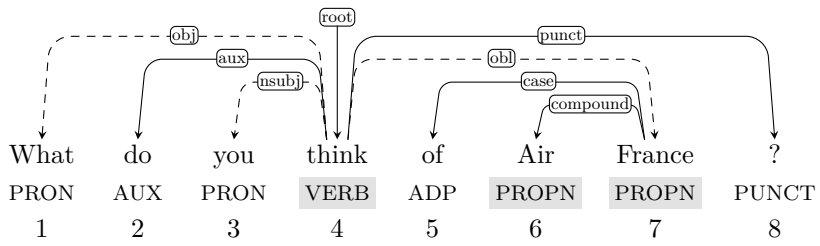
2.1 Glot500-m Model Architecture

We probe Glot500-m, a multilingual transformer language model covering 511, mostly low resource, languages (Imani et al. 2023). Built on top of the XLM-R-base architecture (Conneau et al. 2020), Glot500 uses a transformer with 12 layers, 768 hidden dimensions, and 12 attention heads, for a total of 86M parameters in the transformer components.

The key difference from base architecture is the extended vocabulary generated through continued pre-training with a masked language modeling (MLM) objective on a custom multilingual corpus. This included training a SentencePiece tokenizer on unigram language modeling, where approximately 100K tokens overlapped with XLM-R's original vocabulary, resulting in 151K new tokens. Vocabulary expansion increased the total model size to 395M parameters (compared to XLM-R's 278M), with the difference attributed to the larger embedding matrix. The vocabulary expansion addresses the need to include morphologically rich and typologically diverse languages, making the model well-suited for probing linguistic complexity metrics corresponding to morphosyntactic properties of natural language sentences.

3. Question Type and Linguistic Complexity Signals

We implement a set of probing tasks designed to test whether multilingual models provide better access to sentence-level linguistic properties than traditional representations: question type classification and continuous linguistic complexity score prediction. By comparing the performance of supervised learning algorithms trained on different features at classification and regression tasks, we can determine whether contextualized sentence embeddings capture abstract linguistic patterns better than term frequency features (TF-IDF). We assume that algorithms and probes that demonstrate better ability to recover linguistic labels beyond surface statistical correlations possess stronger linguistic generalization capabilities. This comparative framework allows us to qualify absolute performance differences, but also to measure the relative expressivity of different sentence representation approaches. Specifically, we can investigate which linguistic information is captured in each representation space, by comparing the gap between algorithms trained using TF-IDF features and probes trained on frozen Glot500-m (Imani et al. 2023) contextual embeddings.



Token	root	nsubj	aux	obj	obl	punct	case	compound
Tree Depth	0	1	1	1	1	1	2	2
Dependency Length	0	1	2	3	3	4	2	1

Figure 1: Dependency parse diagram showing grammatical relations, token indices (1-8) and POS tags, with dashed arcs highlighting verbal edges, function words in gray and a table listing dependency distance and tree depth.

Token Count is a straightforward way to measure sentence complexity. It simply refers to the number of processed segments in a sentence, |T|. In languages like English, tokens are words and punctuation marks. In Japanese or Korean, which do not use spaces between words, tokens are aligned with grammatical morphemes rather than orthographic words. Generally, the more tokens a sentence has, the more likely it is to require greater processing efforts (Iavarone et al. 2021).

Lexical density is the ratio of content words (nouns, verbs, adjectives, adverbs) to the total number of tokens excluding punctuation. This metric captures the information density of a sentence and often serves as evidence of register difficulty due to its variation across domains.

$$\text{LD} = \frac{|\text{content words}|}{|\text{T}| - |\text{PUNCT}|} = \frac{3}{7} = 0.428 \quad (1)$$

Average Dependency Length is the linear distance between words and their syntactic heads, across all dependency links in a sentence. This measure directly reflects cognitive processing load, as longer dependencies require holding more information during processing. Futrell et al. (2015) provide compelling evidence in 37 languages, showing that all human languages maintain shorter dependency lengths than would occur by random chance.

$$\text{ADL} = \tfrac{1}{N-1} \sum_{\text{token}(1)}^{N-1} |\text{dep}(i) - \text{head}(i)| = \tfrac{12}{6} = 2 \quad (2)$$

Where N is the number of tokens (i.e., words) without any punctuation and excluding the root of the sentence $(N-1)$.

Maximum Tree Depth measures the longest path from root to leaf in a dependency structure, revealing how deeply embedded linguistic elements are within a sentence. For token index i in sentence S, Max Depth is defined as:

$$\text{MTD} = \max_{\text{token}(i) \in S} \text{Depth}(i) = 2 \quad (3)$$

Average Verbal Edges, sometimes called verbal arity, is a measure of direct dependents (arguments and modifiers) attached to each verb in a sentence. In a dependency structure, these correspond to edges from a verb to its governing words, such as objects, subjects, or adjuncts, but excluding punctuation and auxiliary verbs (Brunato et al. 2020).

$$\overline{\text{VE}} = \frac{1}{|\text{VERBS}|} \sum_{v \in \text{VERBS}} \text{dependent}(v) = 3 \quad (4)$$

Average Subordinate Chain Length is calculated as the ratio of the combined length of all subordinate clauses and the total number of clauses in a sentence. It reflects the level of propositional embedding and recursion. Although the dependency structure in Figure 1 contains no subordination, it remains crucial for capturing the clausal hierarchy of nested sentences.

$$\text{ASC} = \frac{\text{sum of sub. chain lengths}}{\text{number of sub. chains}} = 0 \quad (5)$$

3.1 Question Type Classification Task

Classifying interrogative sentences as either polar (yes-no) questions or content (wh-) questions is one of the fundamental distinctions in linguistic typology. This contrast is considered a near-universal across the world's languages, despite substantial differences in how these question types are marked. We investigate whether sentence representations derived from multilingual language models encode signals robust enough to distinguish between these fundamental question types across different languages. We structure this as a straightforward binary classification problem where models must categorize questions as either polar (e.g., 'Is it raining?') or content-based (e.g., 'Where is the station?'). We compare two-layer MLP probes trained on contextualized embeddings from a frozen multilingual encoder against classical baseline predictors trained on surface-level features (TF-IDF), using selectivity controls to measure whether base performance metrics reflect genuine encoding of linguistic structure or superficial correlations in the data.

Forming question sentences involves several grammatical mechanisms. For example, in English and other SVO languages, questions often involve placing a specific lexical item (wh-word) at the beginning of a sentence (e.g., 'What did John buy?'), while in Japanese and other SOV languages, wh-words typically do not move positions (e.g., 'John-wa nani-o kaimashita ka?' [John-TOP what-ACC bought Q]) (Dryer 2013b). Similarly, polar questions often vary in basic grammatical profiles: English uses subject-auxiliary inversion (e.g., 'Did you eat?'), Russian uses the interrogative particle 'li' (e.g., 'Slishal li ty novosti?' [Listen Q you-NOM news-ACC]), while Finnish mainly uses the enclitic particle '-ko/-kö' attached to the finite verb (e.g., 'Söitkö?' [Ate-Q (you-NOM)]) cf. (Shopen 2007, Dryer 2013a).

Moreover, languages use different strategies to directly mark question categories, ranging from explicit to relatively implicit systems (Dryer 2013a). Explicit marking languages use dedicated particles (e.g., Arabic 'هل' for polar questions), specific suffixes (e.g., Finnish '-ko/-kö'), or consistent grammatical transformations like the English auxiliary inversion that clearly signal question type. In contrast, languages with implicit strategies rely more on context, prosody, or ambiguous markers that serve multiple grammatical functions. Japanese and Korean, for example, use particles that mark all question types with few distinctions, while Indonesian relies more on optional markers and word order variations that can introduce ambiguity.

Cross-linguistic diversity makes question type a relevant category for multilingual probing. From a processing perspective, success in type classification requires recognizing both overt markers (wh-words, question particles) and their structural role within the broader syntax. While lexical cues like 'what' or 'where' might appear as straightforward signals for content questions, they frequently appear in declarative constructions that would confuse purely statistical models. For instance, in English free relative clauses (e.g., 'I'll eat what you cook') and indirect questions (e.g., 'She asked where the station is') wh-words serve functions entirely different from posing questions. Similarly, in Finnish, the interrogative particle that typically marks polar questions can also appear in conditional clauses (e.g., 'Tulisitko, lähtisin kanssasi' [Come-COND-Q], [go-COND I with you-NOM]).

These patterns illustrate why successful question type classification potentially requires sensitivity to both lexical and structural cues. The extent to which different representation types capture these patterns across different languages remains an open empirical question. Therefore, while we hypothesize that contextual embeddings capture such cross-linguistic generalizations more effectively than frequency-based features, we investigate this using controlled comparisons across languages and encoder layers.

3.2 Linguistic Complexity Metrics

The task of linguistic complexity prediction addresses continuous numerical properties of sentences. It allows us to ask whether multilingual representations encode aspects of syntactic and lexical organization that reflect established linguistic predictors of processing difficulty. Unlike categorical distinctions, structural complexity is not always easily reducible to surface cues or lexical items. We define complexity prediction as a regression task:

$$f_{\text{complexity}} : X \rightarrow \mathbb{R}^{+} \tag{6}$$

where X is an input question representation, and the output is a positive scalar complexity score. This framing allows us to assess how different encoding strategies capture the structural properties that contribute to sentence processing load.

While linguistic complexity is studied across disciplines like cognitive science, applied linguistics, and language typology, we focus on approaches that combine theoretical and computational methods. For example, recent work used LLMs to predict reading times, dependency parsing to evaluate syntactic complexity, and applied machine learning techniques to automatically assess text readability across languages (Brunato et al. 2018, Sarti et al. 2021).

From a functional perspective, abstract complexity emerges at the intersection of processing constraints, communication pressures, and structural variation. Languages balance expressive power with learnability and efficiency through different grammatical strategies (Sinnemäki 2011). This balance spans different linguistic systems: agglutinative languages may use multiple morphemes where each corresponds to a single form, while fusional languages use fewer morphemes that combine multiple features simultaneously (Gibson et al. 2019). Similar patterns appear throughout linguistic structure, where increased efficiency in one dimension (e.g., fewer morphemes lead to shorter words) often corresponds to increased complexity in another (e.g., case-ending ambiguity).

Measuring linguistic complexity usually draws on processing theories of grammar (Hawkins 2007), which investigate whether grammars tend to minimize domains of linguistic dependencies. While these theories were originally developed to model human sentence processing, we can operationalize similar principles computationally to create quantitative proxies for linguistic complexity. For example, Futrell et al. (2015) demonstrated cross-linguistic tendencies toward dependency length minimization, providing computational evidence for efficiency patterns in syntactic structure.

Our complexity metrics target structural properties that correlate with increased processing load across languages. Rather than directly modeling human cognitive processing, we use these metrics as computational proxies to investigate whether multilingual representations encode structural information that aligns with established linguistic complexity dimensions. If multilingual models can recover these complexity patterns, this would suggest that such structural information is encoded in their learned representations, providing insight into what linguistic knowledge these models capture.

This alignment between neural representation and computational-linguistic theory provides a principled way to assess the linguistic knowledge captured by multilingual encoders. Since linguistic complexity lacks a standardized model, we define the working notion as an unweighted sum of multiple facets of processing difficulty, each capturing a different aspect of sentence structure. Figure 1 shows a parsed example of a content question from the English EWT treebank (Nivre et al. 2020) annotated with the information required to compute the linguistic profile (Brunato et al. 2020).

The chosen submetrics correspond to findings related to complexity trade-offs in grammars. Sinnemäki (2011) and Futrell et al. (2015) argued that languages exhibit systematic compensation patterns between different complexity dimensions. By training regression algorithms on both composite complexity scores and individual submetrics, we can identify which dimensions of complexity

are most accessible in sentence representations and how this accessibility varies across layers of the model and languages seen during pre-training. Deeper trees, longer dependencies, and longer subordination chains have all been shown to contribute to processing difficulty (Futrell et al. 2015, Brunato et al. 2018, Miaschi et al. 2024).

The tendency of different languages to mark sentence mood via word order, particles, or auxiliary inversion aligns with well-documented efficiency principles in grammar. For example, Hawkins' *Minimize Domains* principle predicts that fronted lexical items in content questions reduce processing effort by allowing early identification of clause type (Hawkins 2007). While our task focuses on categorical distinctions, these design principles anticipate the structural variation we investigate in more detail with complexity prediction tasks.

Our fine-grained approach to complexity prediction has broader implications for NLP applications. Syntactic complexity often varies systematically across domains and genres, with academic and legal texts featuring higher degrees of embedding and longer dependencies than conversational language (Brunato et al. 2018). Models that robustly encode these structural dimensions may better adapt to domain shifts. Zhu et al. (2022) have shown that probing for structural linguistic features can predict fine-tuning performance on downstream tasks. This suggests that models with stronger encoding of features related to complexity may generalize better to new tasks. Similarly, Waldis et al. (2025) demonstrated connections between structural probing and safety alignment, with models that better represent syntactic complexity showing reduced tendency to generate toxic content.

3.3 QTC Dataset

In order to carry out our experiments, we designed the Question Type and Complexity (QTC) dataset containing $\sim$ 9000 questions in Arabic, English, Finnish, Indonesian, Japanese, Korean, and Russian, annotated for both type and complexity metrics. The training split is taken from the TyDiQA-GoldP dataset (Clark et al. 2020) and the test split is based on Universal Dependencies (UD) treebanks (Nivre et al. 2020). For the UD portion of our dataset, we selected one treebank per language based on accessibility (no paywall) and absolute information ranking, which corresponds to parser-informativeness scores as proposed and surveyed by Kulmizev and Nivre (2023).

Language (UD Treebank)	Dataset %	Polar %	Content %	Avg. Complexity
Arabic (PADT)	12.9	48.3	51.7	1.55
English (EWT)	15.9	50.0	50.0	1.61
Finnish (TDT)	15.9	49.9	50.1	1.40
Indonesian (GSD)	13.2	48.2	51.8	1.88
Japanese (PUD)	15.4	50.8	49.2	1.66
Korean (Kaist)	10.7	46.9	53.1	1.98
Russian (Taiga)	16.0	50.0	50.0	1.76

Table 1: QTC dataset statistics by language and Universal Dependencies treebank source.

We annotated all questions in the dataset with binary labels indicating question type. For UD data, labeling was performed using a rule-based system informed by interrogative particles, wh-words, and morphosyntactic cues. For example, polar questions were identified using markers such as sentence-final particles in Japanese and Korean or auxiliary fronting in English, while content questions were detected using wh-phrases appropriate to each language. TyDi data comes annotated with question type labels by native speakers, and we manually verified a sample of cases to ensure consistency with our classification rules. Each sentence is also annotated with a composite score based on six normalized submetrics (see Figure 1) extracted using the Profiling-UD framework (Brunato et al. 2020). All features are computed from dependency parses and scaled to the [0,

1] range within each language to support easier comparison. For UD data, we use the original conllu files, while TyDi examples are parsed using UDPipe 2.0 models (Straka 2018) trained on corresponding treebanks.

The full dataset is released with standardized train, development, and test splits, and includes additional variants for control tasks. Each configuration (base, control 1, control 2, and control 3) contains records with sentence text, language code, binary question type, six normalized complexity features, and an unweighted composite score.

3.4 Task selectivity measurements

Following Hewitt and Liang (2019), we created control tasks to measure probe selectivity by generating variants of our dataset with randomly shuffled labels. For each task (question type classification, composite score, and submetric regression), we provide three random permutations of labels that maintain the original distributions but destroy the relationship between text features and target values. The central insight behind this approach is that a diagnostic model should detect linguistically meaningful patterns rather than arbitrary correlations.

Capable probes achieve high performance on genuine labels but low performance on controls, demonstrating selectivity for linguistically relevant information (Hewitt and Liang 2019). This relationship is critical for picking out cases where neural representations encode hypothesized properties and where they enable effective retrieval through other means. We define the relevant transformations with two complementary metrics for each task:

Metrics	Classification	Regression
Absolute Selectivity	$\text{Accuracy}_{\text{linguistic}} - \text{Accuracy}_{\text{control}}$	$\text{MSE}_{\text{control}} - \text{MSE}_{\text{linguistic}}$
Normalized Selectivity	$\frac{\text{Accuracy}_{\text{linguistic}} - \text{Accuracy}_{\text{control}}}{\text{Accuracy}_{\text{control}}}$	$\frac{\text{MSE}_{\text{control}} - \text{MSE}_{\text{linguistic}}}{\text{MSE}_{\text{control}}}$

Table 2: Selectivity Metrics for Classification and Regression Tasks

We evaluate performance using accuracy across balanced binary classes for question type classification, with raw deltas representing absolute selectivity. Normalized selectivity expresses the relative improvement over the control task. For example, a model achieving linguistic task accuracy of 0.92 and mean control task accuracy of 0.65 gets the absolute score of 0.27 and a normalized score of 0.42, corresponding to a 42 % improvement.

For complexity and submetric regression tasks, we report mean squared error (MSE) as a primary metric. Since all numerical labels are normalized within each language, even modest differences may be indicators of meaningful performance gain. Here, a positive absolute selectivity indicates better performance on the linguistic task, while normalized scores represent the percentage reduction in error compared to controls. As before, an MSE score of 0.08 on real and 0.12 on control labels results in 0.04 on the absolute scale, but corresponds to a 33% gain on the normalized scale.

We apply this framework to two types of models: classical regressors trained on subword TF-IDF features, and neural probes trained on contextual embeddings from the Glot500-base multilingual encoder. The TF-IDF models serve as baselines for what can be extracted from word level statistics alone. We evaluate both linear (Ridge and Logistic Regression) and nonlinear (XGBoost) models, as each may capture different structural signals.

For contextual embeddings, we use two-layer MLP probes trained on frozen sentence representations extracted from all 12 encoder layers. We systematically evaluate probes trained on representations from each layer (layers 1-12) to identify where different linguistic properties are most effectively encoded, with early layers (1-4), middle layers (5-8), and late layers (9-12) representing different levels of 'specialized abstraction' (Tenney et al. 2019a, Jawahar et al. 2019a, Rogers et al. 2020).

This evaluation framework is not designed to maximize performance, but to diagnose where, when, and how well sentence-level linguistic structure is recoverable from different representations. While contextual embeddings offer richer representational capacity, our experiments do not assume that they consistently outperform TF-IDF models. In fact, we observe that performance varies by task, submetric, and language. Moreover, occasional nontrivial performance on control tasks suggests that not all learned structure reflects true linguistic generalization. For this reason, we interpret all results through the lens of selectivity, focusing on relative gains rather than absolute accuracy alone. In the next section, we describe our experimental configurations in detail, including feature extraction, model architecture, and layer-wise probing setup.

4. Experiments

The dataset processing script and experiment runners can be found on GitHub.[3] Our experiments are organized by task and modeling approach, allowing for custom extensions to existing diagnostic models and text representations. We start by presenting baseline experiments using subword TF-IDF features, before moving on to Glot500 feature extraction and MLP probes. We conclude this section with an overview of experimental configurations and implementation details.

4.1 Term Frequency Baselines

In order to establish interpretable baselines, we use linear regression algorithms and decision tree ensembles trained on subword features constructed using the approach described by Wangperawong (2022). This method, explained in the next section, avoids language-specific preprocessing techniques like stemming or stopword removal, while still capturing lexical and morphological variation through Byte Pair Encoding (BPE) subword segmentation (Gage 1994, Sennrich et al. 2016). The choice of frequency-based representations provides a transparent counterpart to the more opaque contextual embeddings used to train our neural probes.

4.1.1 SUBWORD TF-IDF

Each sentence in the dataset is tokenized using the subword tokenizer from the M2M-100[4] model, which results in a combined vocabulary of approximately 128,000 multilingual subword units. This large vocabulary ensures comprehensive coverage across our language set, with roughly $15,000 - 20,000$ subword tokens per language, plus shared tokens that appear across multiple languages. The tokenizer uses a special underscore symbol (_) to mark the beginning of words and distinguish between subwords that occur at word boundaries and those that appear internally. For any segmented input not covered by the vocabulary, it assigns a special <UNK> token to indicate an unknown or out-of-vocabulary unit.

We generate sparse vectors from tokenized input sentences using the subword TF-IDF weighting schema shown in Equation 7, followed by cosine normalization to ensure comparable vector shapes.

3. `https://github.com/rokokot/qtype-eval`
4. https://hf.co/m2m_100

The final result is a high-dimensional sparse vector for each question, with dimensionality equal to the vocabulary size and non-zero values indicating subword importance.

$$\text{weight}(i,j) = \begin{cases} \frac{n_{i,j}}{\sum_{i' \in j} n_{i',j}} \cdot \left(\log \left(\frac{N}{\text{df}_i + 1} \right) + 1 \right) & \text{if } n_{i,j} > 0 \\ 0 & \text{if } n_{i,j} = 0 \end{cases} \tag{7}$$

$n_{i,j}$	raw count of token i in document j	$\sum_{i' \in j} n_{i',j}$	total number of tokens in document j
N	total number of documents in the collection		
df_i	number of documents containing token i	$\frac{n_{i,j}}{\sum_{i' \in j} n_{i',j}}$	normalized term frequency

cf. (Manning and Schütze 1999)

While neural word embeddings like Word2Vec or fastText offer alternative baseline approaches, we chose subword TF-IDF specifically because: 1) unlike static embeddings, it captures document statistics such as term frequency patterns, inverse document frequency weights, and vocabulary distribution, 2) it requires no pre-training, making it more appropriate as a lightweight baseline, and 3) it provides a more interpretable signal where feature importance directly corresponds to subword frequency statistics.

Unlike traditional word-level TF-IDF, which requires language-specific stemming and stopword removal, subword tokenization often preserves morphological information that might otherwise be lost with standard word tokenization. For languages in our dataset with relatively more complex morphology (Finnish, Russian, Arabic), this approach preserves important affixal and inflectional information that would be lost with standard word tokenization. For example, in Russian, the interrogative particle 'ли' (li) is a canonical mark of polar forms which our system tokenizes with the subword '_ли'. Similarly, in English, auxiliary verbs at the beginning of sentences strongly signal polar questions. In the question 'Is it raining outside?', the sentence-initial auxiliary 'is' creates a yes/no question structure that our models learns to recognize through frequency patterns in the training data.

4.1.2 LINEAR PREDICTORS

Although TF-IDF cannot capture hierarchical structure directly, it can detect the lexical signals that mark such structures, namely complementizers, relative pronouns, and specialized punctuation. For highly inflected languages, these markers often appear as distinct subword units within our representation space. If the pattern is regular, it allows us to identify properties we connect to processing complexity without explicitly parsing the sentences. For example, subordinating conjunctions like 'because' (tokenized as '_be', 'cause') signal embedded clauses, while relative pronouns like 'which' and 'that' indicate modification structures. Similarly, temporal markers like '_When' and quantity indicators like Russian '_Ско' (how many/much) also show distribution patterns in our corpus that correlate with specific complexity dimensions, particularly in sentences involving subordination or quantified expressions. Our regression models detect these patterns in subword distribution and use them to predict complexity metrics.

To evaluate how effectively subword features capture linguistic signals, we train several machine learning models of varying complexity. As performance baselines, we implement scikit-learn (Pedregosa et al. 2011) constant predictors that output either the majority class (for classification) or the mean value (for regression) regardless of input features. These baseline predictors establish the minimum performance threshold that any informative model should exceed.

For linear modeling, we implement Logistic Regression for the binary classification task of distinguishing polar from content questions. This algorithm computes class probabilities using a sigmoid

function applied to the weighted sum of features, and learns weights that maximize the likelihood of the training data. The resulting weight vector indicates which subword features are most predictive of each question type. For example, we find that interrogative markers consistently receive the highest weights, with Arabic '_هل' and Indonesian '_Apakah' (both translated as the auxiliary verb 'do' in English) showing strong associations with polar questions.

Similarly, our regression tasks use Linear Regression with L2-regularization that minimizes the sum of squared errors (MSE). This regularization is particularly important given the high dimensionality of our feature space (>100,000 dimensions) relative to our training dataset size (approximately 1,500 examples per language). Linear models face known limits that affect our experiments, mainly because they cannot capture important non-linear relationships between linguistic features. Regression models may be unable to predict metrics where features interact in ways that violate linearity assumptions. For instance, a conjunction like 'and' can join either simple noun phrases (low complexity) or entire clauses (high complexity), but a linear model must assign it a single fixed weight regardless of context.

Classification		Regression	
Feature	Importance	Feature	Importance
_هل	0.038	_هل	0.020
_Onko	0.035	_Apakah	0.011
_ли	0.034	_のは	0.010
_Is	0.025	_apakah	0.009
_Apakah	0.024	_luas	0.008
_是	0.023	_ли	0.008
_?	0.023	_Ско	0.007
_何	0.016	_Onko	0.006
_apakah	0.016	_When	0.006
_か	0.013	_身	0.006

Table 3: Top feature importance scores from XGBoost models for classification and complexity regression tasks.

4.1.3 GRADIENT BOOSTED TREES

We capture nonlinear relationships between subword features using XGBoost classification and regression models. Our implementation uses the native XGBoost Python API (Chen and Guestrin 2016) rather than the scikit-learn wrapper to take advantage of custom evaluation metrics. This enables more precise control over the training process and supports detailed performance analysis across languages and tasks.

The models are configured with a maximum decision tree depth of 6 and an ensemble of 200 trees, striking a balance between model complexity and generalization ability. Early stopping with a patience of 10 epochs prevents overfitting while allowing the model to learn meaningful patterns, and for regularization, we apply L1 (alpha=0) and L2 (lambda=1) penalties. We configure instances of the model class for classification tasks, using the logistic loss function with a binary objective or mean squared error as the evaluation metric in regression tasks.

The key advantage of XGBoost over linear models is its ability to discover nonlinear feature interactions automatically through recursive binary splitting. Nonlinearity is particularly valuable for multilingual tasks, where relationships between subword features may vary and require different splitting conditions. This plays a role in the classification of Finnish, Russian, and English questions, where particles, punctuation, and suffixes interact with stems and roots of the sentence in ways that linear models struggle to capture. Decision tree architectures also prove to be effective at handling

classification tasks in languages with less distinct marking patterns in grammar, but some regularity in the interrogative vocabulary.

Table 3 shows top feature importance scores across all languages for question type classification and complexity regression. Across both tasks, interrogative markers receive the highest importance scores, with the Arabic polar question marker '_هل' (hal) ranking first (0.038), followed closely by Finnish '_Onko' (0.035) and Russian '_ли' (li) (0.034). Although partially overlapping with classification, regression tasks show lower overall importance scores, likely due to a wider set of features contributing to predictions. In general, the features deemed important by regression models cannot be categorized as neatly as interrogative ones, and explicit patterns in which subwords contribute to predictions of specific submetrics are somewhat dampened by low importance scores and overall low performance.

4.2 MLP Probes on Glot500 Sentence Embeddings

While subword vectors provide surface-level lexical statistics, contextualized embeddings encode richer semantic and structural information through their learned parameter space. In our probing configuration, summarized in Table 4, we attempt to balance expressivity of probes (the variety of feature relationships they can learn) and constraints (factors that limit their ability to learn tasks independently of the representations). This setup also partly addresses the view that cautions against using "too powerful" probes that aim for expressivity but do not take the constraints into account (Hewitt and Liang 2019). Our probes are more sophisticated than the linear architectures common in earlier work, and side more with the camp arguing in favor of more complex probes for more complex tasks (Conneau and Kiela 2018, Belinkov 2022). While linear probes are simple, they assume that linguistic properties are linearly separable in representation space. The choice of deeper models is motivated by the need to capture potentially nonlinear relationships in multilingual data.

4.2.1 CLASSIFICATION AND REGRESSION PROBES

For each task type, we use a separately configurable multi-layer perception probe. Both probe types process the same mean-pooled sentence representations from each layer, allowing for direct comparisons of where in the model different linguistic properties are most effectively encoded. Identical input representations ensure that performance differences can be attributed to the properties of the layer representations rather than differences in feature extraction.

Classification probes are two-layer MLPs with a hidden size of 385 units. This allows for more expressivity while following the methods appropriate for binary classification, such as the Gaussian Error Linear Unit activation function (Hendrycks and Gimpel 2023) that has been shown to facilitate better learning. We also use a modest dropout rate and a final sigmoid activation function to produce probability outputs between 0 and 1, with values above 0.5 classified as content questions and those below as polar questions. The classification objective is the standard binary cross-entropy loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{m}\sum_{i=1}^{m}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \tag{8}$$

where m is the batch size, y_i the true binary label, and $\hat{y}_i$ the predicted probability.

Regression tasks use a similar probe configuration, only with a deeper three-layer MLP and a smaller dimension of 128 units, which is optimized for continuous prediction tasks. We use the SiLU activation function, supported by some evidence in outperforming ReLU and GELU on regression tasks (Ramachandran et al. 2017). The lower learning rate for regression (1e-4 vs. 1e-3) allows for

finer-grained optimization on the continuous output space. Moreover, regression models receive an additional epoch of patience during early stopping to accommodate the potentially noisier learning curves when predicting continuous values. The objective is the standard mean squared error:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{9}$$

where y_i is the real value and $\hat{y}_i$ the prediction.

It is worth noting that the configurations presented here were determined through targeted grid searches on our development set, and they may not transfer optimally to other datasets or language combinations. Different linguistic phenomena, dataset sizes, and language sets may require different probe configurations to achieve optimal results. Proper tuning remains a critical step in multilingual probing studies to ensure fair and meaningful comparisons, admittedly at some cost to reporting consistency.

Predictor	Classification	Regression
Dummy	Majority class predictor	Mean value predictor
Linear	Logistic regression with C=1.0, solver='liblinear', L2 regularization, and max 1000 iterations	Linear regression with $\alpha = 0.001$ and standard ridge (L2) regularization
XGBoost	XGBoost classifier with 100 estimators, max tree depth of 6, learning rate=0.1, and a 'logloss' eval metric	XGBoost regressor with 100 estimators, max tree depth of 6, learning rate=0.1, and a [mse','rmse', 'mae'] eval metrics
Probe	2-layer MLP with hidden size=384, dropout=0.05, lr=1e-3, activation=gelu, batch size=16, and early stopping (patience=3)	3-layer MLP with hidden size=128, dropout=0.2, lr=1e-4, activation=silu, batch size=16, and early stopping (patience=4)
Finetuned	Glot500-m with a 2 layer head, head (hidden) size=786, dropout=0.05, lr=1e-3, and batch size=16	Glot500-m with a 3 layer head, hidden size=786, dropout=0.2, lr=1e-4, and batch size=16

Table 4: Hyperparameters for our experimental models and probes.

4.2.2 GLOT500 SENTENCE EMBEDDINGS

For each question in our dataset, we tokenize the input using the model's subword tokenizer, pass the tokenized input through the transformer encoder architecture (12 layers, 768 hidden dimension, 12 attention heads), and extract hidden state vectors from all layers of the model. Although most probing approaches follow the standard [CLS] token pooling for sentence-level tasks, we note the known limitations of this strategy in multilingual contexts. Token pooling is a common technique used for aggregating token-level embeddings into sentence representations (Jurafsky and Martin 2025). It relies on a special token ([CLS], <s>) inserted at the start of a sequence and trained to capture sequence-level information through specialized pre-training objectives.

Limitations of this approach are mainly owed to the fact that the [CLS] token is primarily trained for next sentence prediction tasks during pre-training, which is not a primary objective in the Glot500 model or the underlying XLM-R base. Without specific training, the token lacks the representations found in models trained with NSP (Devlin et al. 2019). Mean pooling is an alternative that aggregates token-level embeddings by computing a weighted average across all tokens in a sequence:

$$h_{\text{sent}} = \frac{\sum_{i=1}^{n} a_i h_i}{\sum_{i=1}^{n} a_i} \tag{10}$$

where $h_i \in \mathbb{R}^d$ is the hidden vector for token i in the sequence extracted from a layer of the model and a_i is the value from the attention mask (1 for actual and 0 for padding tokens). n and d stand for the length of the input sequence and the dimension of the hidden state (768 for Glot500-base). The result is a sentence embedding of the same dimensionality as individual token embeddings (Reimers and Gurevych 2019, Litschko et al. 2021).

Probing methods seem divided on whether any pooling strategy uniformly dominates others. Similarly, our experiments suggest that the deciding factor is the choice of the model and performance analysis on chosen tasks. For all of our tasks, mean pooling consistently outperformed token pooling, which is to be expected. We recommend systematic evaluation of multiple pooling strategies when computational resources permit.

4.2.3 GLOT500 ENCODER PROBING

Most probing studies are set up with the goal of investigating the layer-wise distribution of information relevant for the chosen task (Tenney et al. 2019a, Liu et al. 2019a, Şahin et al. 2020). Following these approaches, we look at all 12 layers of Glot500-base to determine where specific linguistic properties are encoded within the network architecture. For each layer of the model, we extract representations from that layer and apply mean pooling to obtain the embeddings. We then train a task probe using these embeddings and evaluate performance on the real test labels and three controls.

Initial observations showed that normalized selectivity for classification tasks typically ranges from 0.2 (for weaker models) to >0.9 (for the best performers), while regression tasks show values from 0.02 to 0.78, with stronger signals having values around 0.6 and 0.5 for each task. Rather than imposing arbitrary thresholds, we rank layers by their selectivity scores and report the worst and best performing layers for each task and language pair (see Table 5 and 6). To visualize the distribution of linguistic information, we plot performance curves for each complexity submetric, displaying raw performance metrics across all 12 layers (see Figure 3). These curves show whether linguistic properties follow the commonly observed pattern of syntactic information concentrating in lower layers and semantic information in higher layers, or if they follow alternative regular distributions.

4.2.4 FINE-TUNED BASE MODEL

For each plot in Figure 3, we include horizontal reference lines for each of the baseline TF-IDF configurations and one line representing the performance of a fine-tuned Glot500 model on the same task and language. The fine-tuned model is trained end-to-end, updating all encoder parameters, and uses larger, more expressive classification and regression heads than the probes. Despite this standard capacity advantage, our results reveal that fine-tuned models sometimes perform worse than probes trained on frozen representations. Recent work by Kumar et al. (2022) shows some evidence of this pattern being the result of a distortion of pre-trained features, resulting in lower scores compared to linear probing. They show that while fine-tuning may offer slight in-distribution gains, it can degrade the generality and robustness of the learned representations by simultaneously updating both the head and lower network layers.

In our context, the effect is even more pronounced, since the complexity metrics and submetrics we evaluate rely on linguistic and structural information distributed across languages and typologies. Our results further align with the observations of Merchant et al. (2020), who find that fine-tuning can make the model 'forget' linguistic competence acquired during pre-training, with the most rapid loss affecting deeper layers and morphosyntactic features most relevant to our submetric regressions. By comparing performance across different submetrics and model configurations, we investigate which linguistic properties are most sensitive to representational changes during fine-tuning, and

whether the tradeoff between model expressivity and representational stability varies by linguistic phenomenon and language.

4.3 Computing Resources

All experiments were conducted on the Flemish Supercomputer Infrastructure wICE cluster [5] using NVIDIA A100 80GB GPUs. Jobs were allocated wall times up to 72 hours, with each probing experiment taking approximately 5-10 minutes of training time, while fine-tuning experiments took one hour per language-task combination. The complete experimental suite (including baselines) involved over 3300 individual runs across 7 languages, 12 transformer layers, and multiple tasks with control conditions. This includes experiments with linear algorithms, ensembles, and the fine-tuned model.

Training was carried out with batch sizes of 16, gradient accumulation steps of 2-4, and automatic mixed precision. All experiments used fixed random seeds to ensure reproducibility. The total computational cost was approximately 80 GPU hours on A100 hardware.

5. VSC (Flemish Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish Government –department EWI. `https://www.vscentrum.be`

		QUESTION TYPE CLASSIFICATION				COMBINED COMPLEXITY REGRESSION			
		$Acc.$	$\overline{control}$	$\Delta Acc.$	$\widehat{S}$	MSE	$\overline{control}$	ΔMSE	$\widehat{S}$
Dummy baseline	ar	71.4	71.4	0	0	0.059	0.059	0	0
	en	50.0	50.0	0	0	0.038	0.039	0	0
	fi	50.0	50.0	0	0	0.040	0.040	0	0
	id	50.0	50.0	0	0	0.040	0.040	0	0
	ja	50.0	50.0	0	0	0.063	0.063	0	0
	ko	50.0	50.0	0	0	0.036	0.036	0	0
	ru	50.0	50.0	0	0	0.059	0.059	0	0
Linear predictors	ar	98.7	51.9	46.8	0.90	0.038	0.065	0.027	0.41
	en	79.1	50.3	28.8	0.57	0.020	0.042	0.022	0.52
	fi	84.5	44.2	40.3	0.91	0.028	0.041	0.013	0.31
	id	57.3	47.3	10.0	0.21	0.021	0.045	0.024	0.53
	ja	65.2	54.0	11.2	0.20	0.035	0.062	0.027	0.43
	ko	66.5	48.5	18.0	0.37	0.019	0.039	0.020	0.51
	ru	80.0	48.8	31.2	0.63	0.026	0.063	0.037	0.58
Gradient boosting	ar	98.7	50.6	48.1	0.95	0.042	0.056	0.014	0.25
	en	73.6	50.9	22.7	0.44	0.015	0.040	0.025	0.62
	fi	80.9	46.1	34.8	0.75	0.019	0.040	0.021	0.52
	id	69.1	50.0	19.1	0.38	0.019	0.040	0.021	0.52
	ja	64.1	52.5	11.6	0.22	0.028	0.059	0.031	0.52
	ko	66.4	51.2	15.2	0.29	0.015	0.037	0.022	0.59
	ru	87.2	57.5	29.7	0.51	0.024	0.060	0.036	0.60
Optimal Probe	ar	85.7 (2)	71.4	14.3	0.20	0.030 (4)	0.067	0.037	0.55
	en	97.3 (5)	50.0	47.3	0.95	0.017 (1)	0.048	0.031	0.64
	fi	94.5 (5)	50.0	44.5	0.89	0.025 (1)	0.050	0.025	0.50
	id	80.9 (6)	50.0	30.9	0.62	0.024 (4)	0.047	0.023	0.49
	ja	82.6 (10)	39.8	42.8	1.07	0.016 (4)	0.073	0.057	0.78
	ko	76.4 (9)	50.0	26.4	0.53	0.043 (3)	0.093	0.050	0.53
	ru	97.3 (11)	50.0	47.3	0.95	0.039 (6)	0.069	0.030	0.43
Weakest Probe	ar	62.5 (12)	71.4	8.9	0.12	0.057 (12)	0.061	0.004	0.07
	en	89.4 (2)	50.0	39.4	0.79	0.043 (12)	0.045	0.002	0.04
	fi	83.6 (3)	50.0	33.6	0.67	0.042 (12)	0.043	0.001	0.03
	id	62.7 (12)	50.0	12.7	0.25	0.042 (12)	0.046	0.004	0.09
	ja	63.0 (5)	40.2	22.8	0.57	0.058 (12)	0.061	0.003	0.05
	ko	63.6 (1)	50.0	13.6	0.27	0.041 (2)	0.053	0.012	0.22
	ru	90.0 (2)	50.0	40.0	0.80	0.067 (10)	0.075	0.008	0.11
Fine-tuned Glot500	ar	74.1	-	-	-	0.042	-	-	-
	en	91.8	-	-	-	0.020	-	-	-
	fi	92.3	-	-	-	0.030	-	-	-
	id	73.6	-	-	-	0.030	-	-	-
	ja	88.0	-	-	-	0.029	-	-	-
	ko	91.1	-	-	-	0.031	-	-	-
	ru	96.4	-	-	-	0.045	-	-	-

Table 5: Performance metrics for question type classification (accuracy) and combined complexity regression (MSE) tasks across seven languages.

		DEPENDENCY LENGTH		MAX. TREE DEPTH		SUB.CHAIN LENGTH		VERBAL EDGES		LEXICAL DENSITY		N TOKENS	
		MSE	$\widehat{S}$	MSE	$\widehat{S}$	MSE	$\widehat{S}$	MSE	$\widehat{S}$	MSE	$\widehat{S}$	MSE	$\widehat{S}$
Dummy baseline	ar	0.064	0	0.052	0	0.066	0	0.098	0	0.073	0	0.078	0
	en	0.012	0	0.029	0	0.025	0	0.060	0	0.035	0	0.023	0
	fi	0.020	0	0.035	0	0.047	0	0.075	0	0.049	0	0.022	0
	id	0.034	0	0.033	0	0.071	0	0.081	0	0.053	0	0.030	0
	ja	0.096	0	0.094	0	0.131	0	0.102	0	0.063	0	0.074	0
	ko	0.027	0	0.035	0	0.078	0	0.035	0	0.040	0	0.047	0
	ru	0.022	0	0.025	0	0.044	0	0.053	0	0.045	0	0.017	0
Linear predictors	ar	0.040	0.39	0.028	0.48	0.049	0.38	0.091	0.28	0.068	0.04	0.038	0.53
	en	0.006	0.45	0.013	0.60	0.015	0.48	0.045	0.32	0.024	0.18	0.027	0.11
	fi	0.018	0.25	0.030	0.15	0.037	0.30	0.066	0.25	0.058	-0.04	0.015	0.21
	id	0.026	0.27	0.018	0.47	0.035	0.53	0.070	0.07	0.038	0.43	0.022	0.37
	ja	0.076	0.29	0.053	0.37	0.070	0.37	0.094	0.12	0.077	-0.21	0.035	0.43
	ko	0.020	0.26	0.022	0.45	0.045	0.34	0.038	0.09	0.059	0.27	0.046	0.35
	ru	0.019	0.11	0.014	0.48	0.032	0.43	0.042	0.18	0.043	0.33	0.012	0.26
Gradient boosting	ar	0.045	0.22	0.028	0.44	0.057	0.16	0.107	0.13	0.067	0.01	0.038	0.49
	en	0.009	0.21	0.012	0.61	0.016	0.43	0.050	0.21	0.028	0.12	0.018	0.29
	fi	0.022	0.03	0.017	0.52	0.039	0.27	0.066	0.22	0.058	-0.05	0.011	0.37
	id	0.024	0.32	0.029	0.07	0.044	0.39	0.077	-0.06	0.040	0.39	0.035	-0.13
	ja	0.099	0.06	0.063	0.29	0.072	0.36	0.097	0.11	0.078	-0.19	0.028	0.56
	ko	0.018	0.36	0.018	0.53	0.055	0.18	0.037	0.04	0.063	0.22	0.033	0.51
	ru	0.023	-0.07	0.015	0.41	0.041	0.25	0.045	0.06	0.052	0.14	0.009	0.39
Optimal Probe	ar	0.045 (6)	0.42	0.028 (6)	0.48	0.069 (6)	0.12	0.103 (1)	0.23	0.054 (3)	0.10	0.037 (4)	0.59
	en	0.090 (6)	0.23	0.016 (8)	0.46	0.022 (6)	0.25	0.043 (1)	0.35	0.025 (6)	0.11	0.010 (3)	0.63
	fi	0.025 (8)	0.18	0.016 (7)	0.52	0.047 (12)	-0.01	0.080 (12)	-0.01	0.039 (2)	0.14	0.006 (1)	0.59
	id	0.030 (12)	0	0.024 (1)	0.34	0.049 (1)	0.46	0.046 (9)	0.35	0.036 (3)	0.21	0.025 (3)	0.32
	ja	0.087 (1)	0.10	0.072 (9)	0.10	0.053 (6)	0.47	0.007 (6)	0.40	0.071 (7)	0	0.023 (5)	0.66
	ko	0.023 (8)	0.18	0.020 (2)	0.57	0.047 (5)	0.26	0.034 (12)	0	0.062 (8)	0.13	0.073 (9)	0.18
	ru	0.013 (4)	0.29	0.016 (6)	0.41	0.049 (5)	0.14	0.041 (9)	0.18	0.037 (7)	0.14	0.004 (5)	0.68
Weakest Probe	ar	0.073 (12)	-0.13	0.053 (12)	0.05	0.080 (12)	-0.04	0.140 (7)	0	0.065 (5)	-0.02	0.079 (12)	0
	en	0.015 (10)	-0.29	0.029 (12)	0.06	0.031 (3)	0.06	0.076 (8)	0.04	0.030 (12)	0.02	0.023 (12)	0.17
	fi	0.030 (2)	0.07	0.037 (12)	-0.05	0.073 (10)	0.08	0.112 (5)	-0.13	0.051 (12)	-0.03	0.016 (12)	-0.03
	id	0.049 (11)	0.16	0.033 (12)	0.05	0.081 (11)	0.06	0.062 (12)	0.04	0.046 (12)	0.04	0.034 (7)	0.08
	ja	0.122 (3)	0.06	0.103 (10)	-0.10	0.094 (12)	-0.03	0.094 (12)	0.06	0.093 (10)	-0.14	0.063 (12)	0.03
	ko	0.042 (3)	0.18	0.037 (12)	0.02	0.070 (7)	-0.05	0.036 (11)	-0.04	0.074 (6)	-0.07	0.104 (3)	0.12
	ru	0.020 (11)	0.07	0.028 (12)	0	0.058 (6)	0.03	0.057 (8)	-0.08	0.051 (12)	-0.07	0.016 (12)	-0.09
Fine-tuned Glot500	ar	0.056	-	0.038	-	0.069	-	0.030	-	0.055	-	0.056	-
	en	0.008	-	0.022	-	0.019	-	0.039	-	0.023	-	0.020	-
	fi	0.002	-	0.023	-	0.045	-	0.079	-	0.044	-	0.011	-
	id	0.031	-	0.028	-	0.043	-	0.045	-	0.038	-	0.027	-
	ja	0.105	-	0.068	-	0.061	-	0.078	-	0.071	-	0.035	-
	ko	0.025	-	0.029	-	0.046	-	0.034	-	0.074	-	0.070	-
	ru	0.015	-	0.017	-	0.046	-	0.038	-	0.043	-	0.006	-

Table 6: Performance metrics for linguistic complexity submetric regression tasks across 7 languages.

5. Results

Tables 5 and 6, and Figures 2 and 3 report results of our experiments with categorical (question type classification) and continuous labels (complexity metrics). We observe noticeable variation in how linguistic information is distributed across transformer layers, but also point out several language-specific trends in task performance. By examining classification accuracy and regression errors alongside selectivity metrics, we identify where contextual embeddings provide meaningful advantages over alternatives and how these patterns change with different linguistic properties.

5.1 Classification Performance

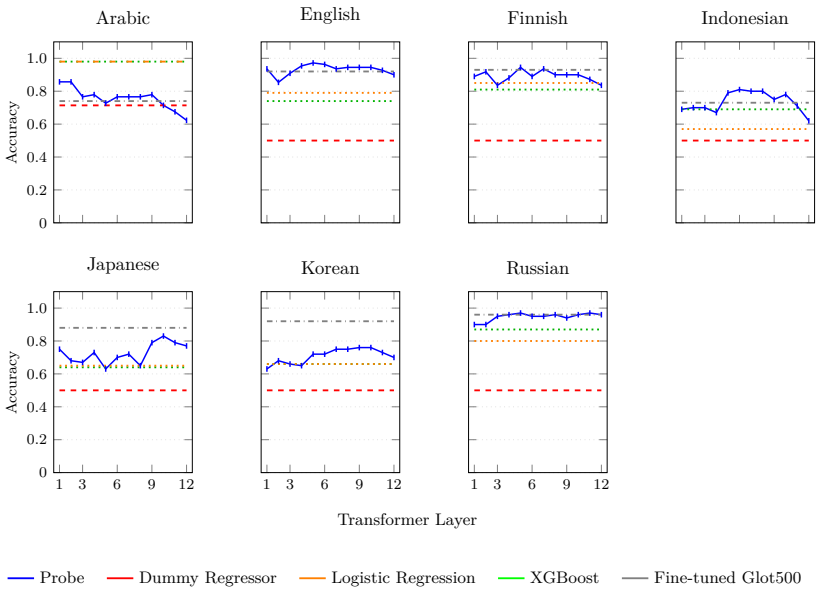


Figure 2: Classification performance metrics across languages and transformer layers

Comparing baseline predictors trained on subword frequency features to diagnostic models trained on representations of a frozen Glot500 encoder reveals that grammar and surface lexical properties play an important role in classification performance. The ability to capture different question types varies across languages and relates to the grammatical elements used to distinguish between categories, especially in languages with regular and unambiguous marking strategies.

Languages with explicit marking strategies demonstrate high selectivity across all predictors. Arabic shows this most clearly with its polar question particle '_هل' (hal), achieving TF-IDF selectivity scores (linear: 0.90, XGBoost: 0.95) that exceed optimal probes (0.20). Russian shows similar patterns with the '_ли' (li) marker, where linear regression reaches 0.63 selectivity and optimal probes achieve 0.95. Finnish polar suffix '-ko/-kö' and auxiliary inversion in English follow similar trends.

In contrast, languages with implicit marking strategies show a clear improvement in selectivity from simpler to more sophisticated predictors and features. Japanese has the largest gap, going from 0.22 (XGBoost) to 1.07 using a probe on the optimal layer (10), where the probe achieves 82.6% accuracy compared to just 39.8% on control tasks. Indonesian follows a similar pattern, with probes reaching 0.62 (6) compared to 0.21 selectivity of linear regression. This pattern highlights the value of contextual embeddings for capturing syntactic variation and subtle structural cues beyond surface markers.

Feature attribution from XGBoost supports this distinction. Table 3 shows that the subwords contributing most to class predictions include Arabic '_هل' (0.038), Finnish '_Onko' (0.035), Russian _ли' (0.034), and English '_Is' (0.025), all canonical markers of polar questions. Conversely, ambiguous markers like '?', Japanese '_か' (ka), or Indonesian '_apakah' receive lower scores while still remaining top contributors. These feature importances hint that linear and gradient boosted predictors effectively exploit explicit markers but struggle with more implicit cues.

As shown in Figure 2, results of fine-tuning the Glot500 model show strong accuracy for most languages, reaching 90% accuracy in English, Finnish, Japanese, Korean, and Russian. However, the relative benefit varies. The largest improvements are observed for Japanese and Korean, which gain 15 and 25 points, respectively, over their best accuracy baselines. These results suggest that fine-tuning increases sensitivity to type cues, especially in languages lacking overt interrogative markers. In contrast, Arabic performance drops significantly compared to its baseline models, with a 24-point gap between the fine-tuned accuracy (74.1%) and that of the baseline classifiers (98.7%).

5.2 Regression Performance

Complexity score regression shows distinct patterns for the combined abstract complexity metric and individual submetrics. Glot500 probes perform well, but with significant variation. Japanese stands out with the highest selectivity of 0.78 at layer 4, which is a substantial increase over ridge (0.43) and XGBoost (0.52). English follows this pattern with probes achieving 0.64 selectivity at layer 1, compared to XGBoost (0.62) and linear models (0.52). Notably, gradient boosting ensembles performs quite well across several languages, achieving selectivity scores between 0.52-0.62 for English, Finnish, Indonesian, Japanese, and reaching 0.6 for Russian. The optimal probe selectivity scores range from 0.43 (Russian) to 0.78 (Japanese), with most languages showing best performance in earlier layers (1-4).

Although contextual embeddings have an advantage, the gap between XGBoost and optimal probes is often modest, indicating that sophisticated TF-IDF models can capture some of the structural information needed for complexity assessment. For instance, in English, XGBoost achieves 0.62 selectivity compared to the optimal probe's 0.64, and in Korean, XGBoost outperforms the optimal probe (0.59 vs. 0.53). This pattern holds across most languages, with Russian showing the largest advantage for gradient boosting (0.6) over its optimal probe (0.43 at layer 6). Probing results show uniform increase only in Japanese and Arabic (increase of 0.26 and 0.30 points respectively). These results suggest that while neural representations offer advantages, statistical models remain competitive for complexity prediction.

Moreover, fine-tuned models consistently underperform optimal probes in combined complexity predictions. In Japanese, the error of the fine-tuned Glot500 model increased by 80% compared to the optimal probe, with similar decrease observed in English, Finnish, and Russian. We note this as the distinction between ability to capture categorical labels well during fine-tuning, while destabilizing performance on continuous labels like complexity scores. Similar to prior observations by Merchant et al. (2020) and Kumar et al. (2022), fine-tuning for numerical targets appears to increase the distortion of pretrained knowledge, particularly for continuous, compositional properties.

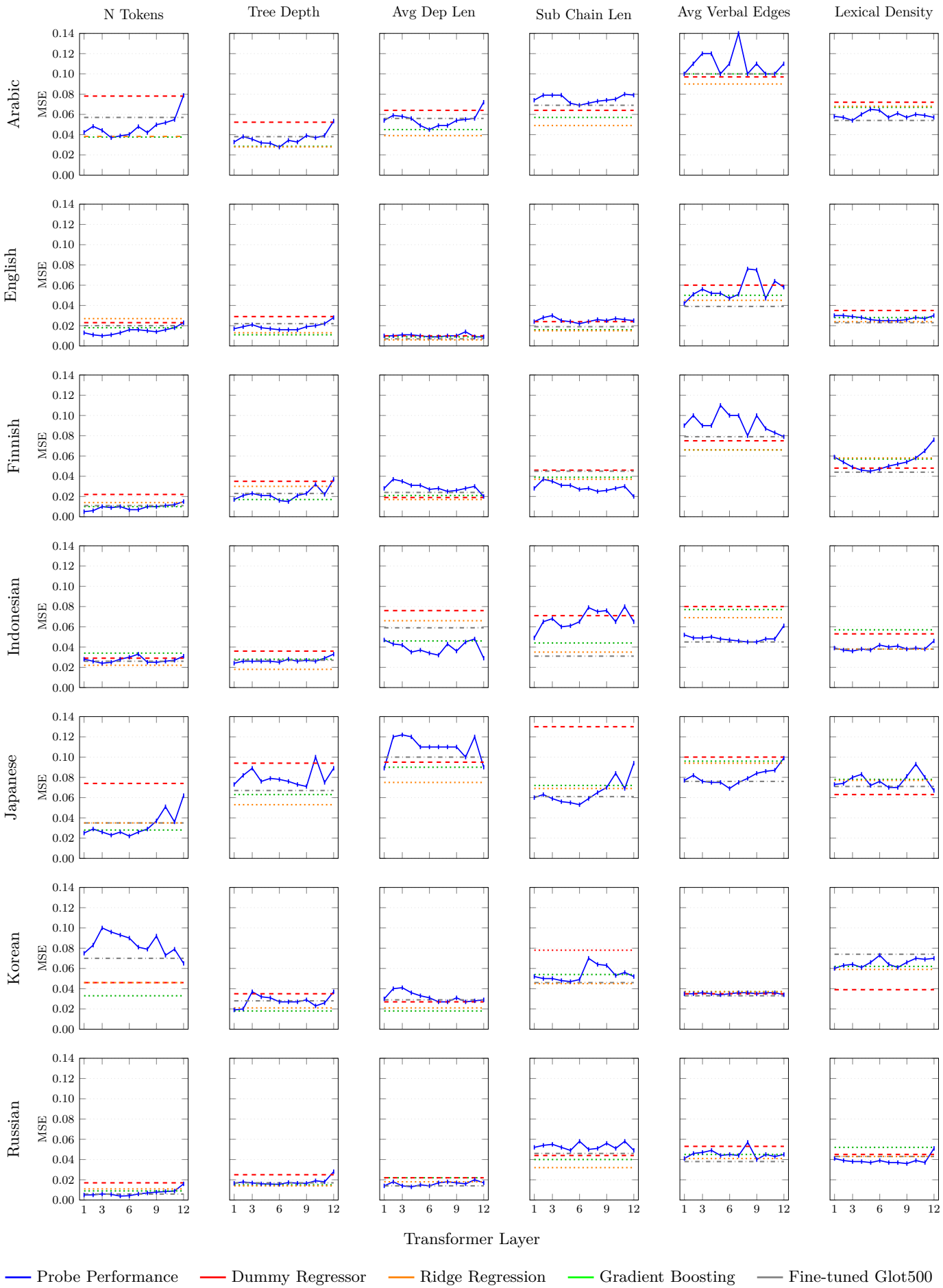


Figure 3: Performance metrics across languages and transformer layers

5.3 Submetric Performance Trends

Individual complexity submetrics show substantial variation across linguistic subsystems (syntax, morphology, and surface properties). Figure 3 provides an overview of how linguistic information is distributed across transformer layers.

Token Count is a seemingly simple metric that measures the number of subword tokens in a sentence as segmented by the respective tokenizers (Glot500's tokenizer for neural probes and M2M-100's tokenizer for TF-IDF baselines). However, performance on different languages varies considerably. In English, probing earlier layers shows clear advantages, with the best probe (layer 3) achieving 0.63 selectivity (MSE: 0.010) compared to linear models (0.11, MSE: 0.027) and XGBoost (0.29, MSE: 0.018). Similarly, both Finnish and Japanese demonstrate improvements in performance when using probes (Finnish: 0.59 at layer 1, MSE: 0.006; Japanese: 0.66 at layer 5, MSE: 0.023).

Overall, line charts highlighting performance at each layer show that lower transformer layers (1-5) usually perform better than deeper layers and statistical baselines at predicting token count. These patterns are particularly strong in Russian (0.68 selectivity at layer 5), Japanese (0.66 at layer 5), and English (0.63 at layer 3). Indonesian probes show more modest improvements (0.32 at layer 3, MSE: 0.025) over linear models (0.37, MSE: 0.022), and Korean probes perform poorly (best selectivity 0.18 at layer 9, MSE: 0.073) compared to baseline models (linear: 0.35, MSE: 0.046; XGBoost: 0.51, MSE: 0.033).

Tree depth performance varies substantially across languages. Finnish probes achieve 0.52 selectivity at layer 7 (MSE: 0.016), Korean 0.57 at layer 2 (MSE: 0.020), and English 0.46 at layer 8 (MSE: 0.016). Linear models consistently outperform or match probes across languages. For English, linear models achieve 0.60 selectivity (MSE: 0.013) compared to 0.46 at the optimal layer. In Arabic, linear models and probes achieve identical selectivity (0.48), with the same MSE (0.028). The layer plots reveal a preference for earlier layers in most languages, with only English, Indonesian, and Russian showing flat trends or slightly increased errors at final layers.

Average dependency length shows moderate performance overall, with the strongest signals in Arabic (0.42 selectivity at layer 6, MSE: 0.045) and English (0.23 at layer 6, MSE: 0.009). Most languages show better performance with linear or XGBoost models than with probes, for example, in English, linear models achieve 0.45 selectivity (MSE: 0.006) compared to the optimal probe with 0.23. English probes exhibit notably flat performance curves that closely track baseline model performance across all layers. Similar flat curves appear in Russian. Japanese and Indonesian are outliers, with linear models at 0.29 selectivity (MSE: 0.076) significantly outperforming probes at all layers for Japanese, while Indonesian optimal probes perform well in terms of error but lack meaningful selectivity.

Overall, subordinate chain length is better captured by linear models than by neural probes. Indonesian shows 0.53 selectivity with linear models (MSE: 0.035) compared to 0.46 with the best probe (layer 1, MSE: 0.049). English shows an even larger gap, with linear models at 0.48 selectivity (MSE: 0.015) versus probes at 0.25 (layer 6, MSE: 0.022). English also displays characteristically flat performance curves across layers for this metric. Only Japanese demonstrates stronger probe performance (0.47 at layer 6, MSE: 0.053) than linear models (0.37, MSE: 0.070).

Verbal arity demonstrates the most extreme variation across languages and layers. Indonesian probes perform well (0.35 selectivity at layer 9, MSE: 0.046) compared to linear models (0.07, MSE: 0.070), with relatively flat performance curves across layers. Japanese similarly shows strong probe performance (0.40 at layer 6, MSE: 0.007) versus linear (0.12, MSE: 0.094). Korean also exhibits notably flat curves, with performance remaining consistent across layers and comparable to baseline models. In contrast, Finnish and Arabic probes perform poorly, with Finnish showing negative

selectivity (-0.01 at layer 12, MSE: 0.080) compared to linear predictors (0.25, MSE: 0.066). This metric displays the most language-specific variation in layer distributions, likely reflecting different grammatical strategies for marking verbs and verbal relationships.

Lexical density consistently shows poor encoding across all configurations. Performance is generally weak across all model types, with selectivity rarely exceeding 0.20. English, Indonesian, and Russian all display flat curves for lexical density that closely track baseline model performance. Linear models outperform probes for Indonesian (0.43, MSE: 0.038) and Korean (0.27, MSE: 0.059). Japanese shows negative selectivity for both linear (-0.21, MSE: 0.077) and XGBoost (-0.19, MSE: 0.078) models, while the best probe achieves zero selectivity (layer 7, MSE: 0.071). Universally weak performance suggests lexical density may require explicit part-of-speech knowledge not adequately represented in any of these representations.

6. Discussion

Our experiments address three questions about multilingual representations: 1) whether contextual embeddings support better generalization than subword frequency features for sentence-level linguistic tasks, 2) how linguistic signal strength varies between worst and best performing layers of the encoder, and 3) whether performance on binary and regression tasks reflects meaningful linguistic generalization. We organize the discussion around these questions, examining performance on classification and complexity tasks across our predictor models, patterns in submetric performance, and addressing selectivity as an approximation of linguistic signal strength.

6.1 Representational Quality of Binary and Continuous Labels

The first question evaluates the capacity of multilingual sentence representations to outperform traditional statistical feature engineering. We observe that contextual embeddings lead to better generalization on question type classification, particularly in languages with implicit marking strategies (e.g., Japanese, Korean, Indonesian). This aligns with claims that pre-trained LLMs effectively capture categorical distinctions when lexical cues are weak or ambiguous (Adi et al. 2017, Jawahar et al. 2019b), but it also speaks in favor of views which hold that distinct grammatical strategies for marking categories make sentence classification easier (Hawkins 2007, Futrell et al. 2015).

Some evidence for the latter idea comes from observations that simple statistical approaches with less sophisticated models perform as well or better in languages with explicit morphosyntactic markers (e.g., Arabic, Finnish, Russian). For example, Japanese probes achieve selectivity scores >1.0 compared to just 0.22 with XGBoost baselines, demonstrating the ability to capture subtle patterns. Similarly, Indonesian probes reach 0.62 selectivity at layer 6, nearly tripling the 0.21 selectivity of linear regression. The pattern confirms that contextual embeddings excel precisely where simple statistical approaches struggle.

Regression tasks equally complicate the view that contextual representations are universally superior. While some submetrics (e.g., token count) benefit from neural representations in early layers, tree depth, subordination, and verbal arity are often better predicted by TF-IDF models. In English, linear models achieve 0.60 selectivity for tree depth compared to 0.46 with the best probe, and 0.48 for subordinate chain length versus 0.25 for the best probe. These results suggest that regression probes do not necessarily encode the targeted structure.

Our observations raise important questions about how we evaluate success in regression probing. Unlike classification, where performance on permuted controls is directly interpretable through clear decision boundaries (a classifier either correctly identifies a question type or not), regression performance is less interpretable. A model might achieve an MSE of 0.016 on tree depth prediction, but

this value alone reveals little about whether the model has learned meaningful structure or merely exploited other patterns. Selective metrics (Hewitt and Liang 2019, Pimentel et al. 2020) quantify the relative gain in performance between linguistic and control tasks. For example, in Japanese complexity prediction, the raw MSE of 0.016 at layer 4 becomes more meaningful when contextualized by the control task MSE of 0.073, yielding a selectivity of 0.78. This approach directly measures a model's ability to distinguish real linguistic patterns from random noise. Even so, we find cases like English tree depth where probes achieve low error (MSE: 0.016) but modest selectivity (0.46), showing that sometimes good performance does not derive from genuine structural encoding.

6.2 Regression Signal Strength

Our second question addressed the issue of where our target structure is encoded in the model. Prior work suggests that lower transformer layers perform better on syntactic targets and higher layers on semantics (Conneau and Kiela 2018, Tenney et al. 2019a, Liu et al. 2019a). Regression results only partially support this view, with early and middle layers often capturing token count and basic structural features like tree depth and subordinate chain length. Performance curves in Figure 3 show two patterns across languages. First, some languages (e.g., English, Indonesian, Russian) display predominantly flat curves for multiple metrics, suggesting relatively uniform encoding of structural information throughout the network. For example, English shows nearly flat performance for dependency length, subordinate chain length, and lexical density, with minimal variation across layers. If we follow conventional wisdom, this could mean that certain structural properties may be encoded as surface-level features that remain equally (in)accessible at all layers.

The second pattern, most evident in Japanese, is that there are no flat curves at all in some cases, partly confirming the idea that linguistic features are more localized and layer-specific. However, most of the trends are flat or only slightly convex. These deserve more attention because they show that most of the submetrics are encoded relatively uniformly across transformer layers rather than concentrating at specific depths. While classification probing often reveals clear transitions between layers specialized for different categorical distinctions, regression probing of continuous linguistic properties presents a different picture. Results suggest that information about dependency length, tree depth, and subordination chains may be represented through distributed mechanisms accessible throughout the network's entire depth. However, they also raise important questions about the actual cause of uniform patterns.

Namely, it is an open question whether similar trends relate to the fact that structural properties we are probing for are encoded through stable features that do not vary across layers, the information is genuinely distributed throughout the network without layer specialization, or our probing methods lack the sensitivity to detect subtle variations in how the signals are represented if there are any signals in the first place.

6.3 Limits of Regression Error Probing

Continuing with the previous discussion, the final question asked whether observed performance reflects a real linguistic signal or mere pattern memorization. Here, regression probing is particularly vulnerable. Classification allows for clear error boundaries and control-based selectivity metrics (e.g., accuracy delta), but regression probes lack standardized interpretive thresholds. A small difference in mean squared error between shuffled and real labels may reflect meaningful structure, noise, or a mix of signal and noise.

Our selectivity analysis provides concrete evidence of this vulnerability. With English tree depth, we observe low MSE (0.016) at later layers but only moderate selectivity (0.46), significantly below the linear baseline (0.60). This indicates that while the probe accurately predicts the metric, it likely

relies on superficial patterns rather than true structural encoding. Similarly, for Finnish verbal edges, we find negative selectivity (-0.01) despite reasonable MSE (0.08), revealing that the probe performs worse than it would on random labels, a sign of overfitting rather than linguistic generalization.

The ranges we observe across submetrics further complicate things. Classification selectivity typically ranges from 0.2 to over 0.9 for strong signals, while regression selectivity spans from near zero to 0.78 at best. This compressed range makes small differences more difficult to interpret without additional context. For example, optimal probe on Russian dependency length shows 0.29 selectivity, which seems weak in classification contexts but potentially represents a meaningful signal in regression.

A certain level of ambiguity is present even in the results of the fully fine-tuned Glot500 model, which performs well in classification but often has larger regression errors than frozen probes. These patterns suggest that fine-tuning, while optimizing for categorical distinctions, may disrupt the structural encodings that represent continuous linguistic properties.

With these observations, we note important challenges in regression probing as an interpretability technique. Without carefully designed baselines and controls, regression results are misleading. Probes may achieve low error while capturing spurious correlations rather than linguistic structure. Unlike classification probes with their clearer performance boundaries, regression probes work with labels where performance metrics require contextual interpretation. They are also prone to scaling effects, as similar error values may represent vastly different levels of performance across different complexity dimensions. Addressing these challenges doesn't diminish the value of regression probes but rather highlights the need for caution. To derive meaningful insights, researchers should employ control tasks with shuffled labels to establish selectivity baselines, careful metric design that isolates more direct signals, and methods to disentangle surface effects from true structure.

7. Conclusion

We introduced a set of methods for evaluating how massively multilingual LLMs encode functional linguistic knowledge related to sentence type and complexity. Rather than measuring general benchmark performance, we conducted probing experiments with selectivity controls that test whether layers of the frozen Glot500-m encoder retain genuine linguistic signals.

Our results show patterns based on typological properties, with subword frequency features outperforming contextual models for languages with explicit interrogative markers. In contrast, contextual embeddings and diagnostic probes prove better for languages with implicit or ambiguous markers, where structural cues cannot be reduced to overt lexical patterns. For complexity regression tasks, we observe fewer regularities. Our probes demonstrate better selectivity in distinguishing genuine linguistic signals from noise, but traditional ML models often outperform neural probes on structural submetrics.

Interestingly, probing frozen representations consistently leads to better results than fine-tuning across multiple tasks. This suggests that fine-tuning degrades structural linguistic representations, which contradicts common assumptions about model optimization. For practical applications, our results suggest that simpler statistical approaches may suffice for certain linguistic phenomena, while contextual embeddings become essential for capturing implicit structural patterns as well as particular submetrics related to morphosyntax. Understanding these trade-offs is crucial for developing more accurate diagnoses of what models know and where they fail.

References

Adi, Yossi, Einat Kermany, Yonatan Belinkov, Ofer Lavi, and Yoav Goldberg (2017), Fine-grained analysis of sentence embeddings using auxiliary prediction tasks, *Proceedings of the 5th International Conference on Learning Representations (ICLR)*. Conference Track. https://openreview.net/forum?id=BJh6Ztuxl.

Beguš, Gašper, Maksymilian Dąbkowski, and Ryan Rhodes (2025), Large linguistic models: Investigating LLMs' metalinguistic abilities. http://arxiv.org/abs/2305.00948.

Belinkov, Yonatan (2022), Probing classifiers: Promises, shortcomings, and advances, **48** (1), pp. 207–219, MIT Press. https://aclanthology.org/2022.cl-1.7/.

Brunato, Dominique, Andrea Cimino, Felice Dell'Orletta, Giulia Venturi, and Simonetta Montemagni (2020), Profiling-UD: a tool for linguistic profiling of texts, *in* Calzolari, Nicoletta, Frédéric Béchet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis, editors, *Proceedings of the Twelfth Language Resources and Evaluation Conference*, European Language Resources Association, pp. 7145–7151. https://aclanthology.org/2020.lrec-1.883/.

Brunato, Dominique, Lorenzo De Mattei, Felice Dell'Orletta, Benedetta Iavarone, and Giulia Venturi (2018), Is this sentence difficult? do you agree?, *in* Riloff, Ellen, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, pp. 2690–2699. https://aclanthology.org/D18-1289/.

Chen, Tianqi and Carlos Guestrin (2016), Xgboost: A scalable tree boosting system, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, Association for Computing Machinery, New York, NY, USA, p. 785–794. https://doi.org/10.1145/2939672.2939785.

Clark, Jonathan H., Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki (2020), TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages, *Transactions of the Association for Computational Linguistics* **8**, pp. 454–470, MIT Press, Cambridge, MA. https://aclanthology.org/2020.tacl-1.30/.

Conneau, Alexis and Douwe Kiela (2018), SentEval: An evaluation toolkit for universal sentence representations, *in* Calzolari, Nicoletta, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Koiti Hasida, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazo, Asuncion Moreno, Jan Odijk, Stelios Piperidis, and Takenobu Tokunaga, editors, *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, European Language Resources Association (ELRA), Miyazaki, Japan. https://aclanthology.org/L18-1269/.

Conneau, Alexis, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni (2018), What you can cram into a single $&!#* vector: Probing sentence embeddings for linguistic properties, *in* Gurevych, Iryna and Yusuke Miyao, editors, *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, pp. 2126–2136. https://aclanthology.org/P18-1198/.

Conneau, Alexis, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov (2020),

Unsupervised cross-lingual representation learning at scale, *in* Jurafsky, Dan, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Online, pp. 8440–8451. https://aclanthology.org/2020.acl-main.747/.

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova (2019), BERT: Pre-training of deep bidirectional transformers for language understanding, *in* Burstein, Jill, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Association for Computational Linguistics, Minneapolis, Minnesota, pp. 4171–4186. https://aclanthology.org/N19-1423/.

Doddapaneni, Sumanth, Gowtham Ramesh, Mitesh Khapra, Anoop Kunchukuttan, and Pratyush Kumar (2025), A primer on pretrained multilingual language models, *ACM Comput. Surv.*, Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3727339.

Dryer, Matthew S. (2013a), Polar questions (v2020.4), *in* Dryer, Matthew S. and Martin Haspelmath, editors, *The World Atlas of Language Structures Online*, Zenodo. https://doi.org/10.5281/zenodo.13950591.

Dryer, Matthew S. (2013b), Position of interrogative phrases in content questions (v2020.4), *in* Dryer, Matthew S. and Martin Haspelmath, editors, *The World Atlas of Language Structures Online*, Zenodo. https://doi.org/10.5281/zenodo.13950591.

Futrell, Richard, Kyle Mahowald, and Edward Gibson (2015), Large-scale evidence of dependency length minimization in 37 languages, *Proceedings of the National Academy of Sciences* **112** (33), pp. 10336–10341, National Academy of Sciences. https://www.pnas.org/doi/abs/10.1073/pnas.1502134112.

Gage, Philip (1994), A new algorithm for data compression, *C Users J.* **12** (2), pp. 23–38, R & D Publications, Inc., USA.

Gibson, Edward, Richard Futrell, Steven P. Piantadosi, Isabelle Dautriche, Kyle Mahowald, Leon Bergen, and Roger Levy (2019), How efficiency shapes human language, Elsevier.

Hawkins, John A. (2007), *Efficiency and complexity in grammars*, Oxford linguistics, repr ed., Oxford University Press.

Hendrycks, Dan and Kevin Gimpel (2023), Gaussian error linear units (gelus). https://arxiv.org/abs/1606.08415.

Hewitt, John and Christopher D. Manning (2019), A structural probe for finding syntax in word representations, *in* Burstein, Jill, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Association for Computational Linguistics, Minneapolis, Minnesota, pp. 4129–4138. https://aclanthology.org/N19-1419/.

Hewitt, John and Percy Liang (2019), Designing and interpreting probes with control tasks, *in* Inui, Kentaro, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Association for Computational Linguistics, Hong Kong, China, pp. 2733–2743. https://aclanthology.org/D19-1275/.

Iavarone, Benedetta, Dominique Brunato, and Felice Dell'Orletta (2021), Sentence complexity in context, *in* Chersoni, Emmanuele, Nora Hollenstein, Cassandra Jacobs, Yohei Oseki, Laurent Prévot, and Enrico Santus, editors, *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*, Association for Computational Linguistics, pp. 186–199. https://aclanthology.org/2021.cmcl-1.23/.

Imani, Ayyoob, Peiqin Lin, Amir Hossein Kargaran, Silvia Severini, Masoud Jalili Sabet, Nora Kassner, Chunlan Ma, Helmut Schmid, André Martins, François Yvon, and Hinrich Schütze (2023), Glot500: Scaling multilingual corpora and language models to 500 languages, *in* Rogers, Anna, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, pp. 1082–1117. https://aclanthology.org/2023.acl-long.61/.

Jawahar, Ganesh, Benoît Sagot, and Djamé Seddah (2019a), What does BERT learn about the structure of language?, *in* Korhonen, Anna, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Florence, Italy, pp. 3651–3657. https://aclanthology.org/P19-1356/.

Jawahar, Ganesh, Benoît Sagot, and Djamé Seddah (2019b), What does BERT learn about the structure of language?, *in* Korhonen, Anna, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, pp. 3651–3657. https://aclanthology.org/P19-1356/.

Jurafsky, Daniel and James H. Martin (2025), *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*, 3rd ed., manuscript. https://web.stanford.edu/ jurafsky/slp3/.

Köhn, Arne (2015), What's in an embedding? analyzing word embeddings through multilingual evaluation, *in* Màrquez, Lluís, Chris Callison-Burch, and Jian Su, editors, *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Lisbon, Portugal, pp. 2067–2073. https://aclanthology.org/D15-1246/.

Kulmizev, Artur and Joakim Nivre (2023), Investigating UD treebanks via dataset difficulty measures, *in* Vlachos, Andreas and Isabelle Augenstein, editors, *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, Association for Computational Linguistics, Dubrovnik, Croatia, pp. 1076–1089. https://aclanthology.org/2023.eacl-main.76/.

Kumar, Ananya, Aditi Raghunathan, Robbie Matthew Jones, Tengyu Ma, and Percy Liang (2022), Fine-tuning can distort pretrained features and underperform out-of-distribution, *International Conference on Learning Representations*. https://openreview.net/forum?id=UYneFzXSJWh.

Litschko, Robert, Ivan Vulić, Simone Paolo Ponzetto, and Goran Glavaš (2021), Evaluating multilingual text encoders for unsupervised cross-lingual retrieval, *Proceedings of ECIR*.

Liu, Nelson F., Matt Gardner, Yonatan Belinkov, Matthew E. Peters, and Noah A. Smith (2019a), Linguistic knowledge and transferability of contextual representations, *in* Burstein, Jill, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Association for Computational Linguistics, Minneapolis, Minnesota, pp. 1073–1094. https://aclanthology.org/N19-1112/.

Liu, Shusen, Zhimin Li, Tao Li, Vivek Srikumar, Valerio Pascucci, and Peer-Timo Bremer (2019b), Nlize: A perturbation-driven visual interrogation tool for analyzing and interpreting natural language inference models, *IEEE Transactions on Visualization and Computer Graphics* **25** (1), pp. 651–660, IEEE Computer Society.

Liu, Yinhan, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, and Luke Zettlemoyer (2020), Multilingual denoising pre-training for neural machine translation, **8**, pp. 726–742, MIT Press. Place: Cambridge, MA. https://aclanthology.org/2020.tacl-1.47/.

Manning, Christopher D. and Hinrich Schütze (1999), *Foundations of Statistical Natural Language Processing*, The MIT Press, Cambridge, Massachusetts. http://nlp.stanford.edu/fsnlp/.

Merchant, Amil, Elahe Rahimtoroghi, Ellie Pavlick, and Ian Tenney (2020), What happens to BERT embeddings during fine-tuning?, *in* Alishahi, Afra, Yonatan Belinkov, Grzegorz Chrupała, Dieuwke Hupkes, Yuval Pinter, and Hassan Sajjad, editors, *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, Association for Computational Linguistics, Online, pp. 33–44. https://aclanthology.org/2020.blackboxnlp-1.4/.

Miaschi, Alessio, Felice Dell'Orletta, and Giulia Venturi (2024), Evaluating large language models via linguistic profiling, *in* Al-Onaizan, Yaser, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, pp. 2835–2848. https://aclanthology.org/2024.emnlp-main.166/.

Muller, Benjamin, Deepanshu Gupta, Jean-Philippe Fauconnier, Siddharth Patwardhan, David Vandyke, and Sachin Agarwal (2023), Languages you know influence those you learn: Impact of language characteristics on multi-lingual text-to-text transfer, *Proceedings of The 1st Transfer Learning for Natural Language Processing Workshop*, PMLR, pp. 88–102. ISSN: 2640-3498. https://proceedings.mlr.press/v203/muller23a.html.

Nivre, Joakim, Marie-Catherine de Marneffe, Filip Ginter, Jan Hajič, Christopher D. Manning, Sampo Pyysalo, Sebastian Schuster, Francis Tyers, and Daniel Zeman (2020), Universal dependencies v2: An evergrowing multilingual treebank collection, *in* Calzolari, Nicoletta, Frédéric Béchet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis, editors, *Proceedings of the Twelfth Language Resources and Evaluation Conference*, European Language Resources Association, pp. 4034–4043. https://aclanthology.org/2020.lrec-1.497/.

Pedregosa, Fabian, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. (2011), Scikit-learn: Machine learning in python, *Journal of machine learning research* **12** (Oct), pp. 2825–2830, MIT Press.

Pimentel, Tiago, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell (2020), Information-theoretic probing for linguistic structure, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, pp. 4609–4622. https://www.aclweb.org/anthology/2020.acl-main.420.

Ramachandran, Prajit, Barret Zoph, and Quoc V. Le (2017), Searching for activation functions. https://arxiv.org/abs/1710.05941.

Ravishankar, Vinit, Lilja Øvrelid, and Erik Velldal (2019a), Probing multilingual sentence representations with x-probe, *in* Augenstein, Isabelle, Spandana Gella, Sebastian Ruder, Katharina Kann, Burcu Can, Johannes Welbl, Alexis Conneau, Xiang Ren, and Marek Rei, editors, *Proceedings of the 4th Workshop on Representation Learning for NLP (RepL4NLP-2019)*, Association for Computational Linguistics, pp. 156–168. https://aclanthology.org/W19-4318/.

Ravishankar, Vinit, Memduh Gökırmak, Lilja Øvrelid, and Erik Velldal (2019b), Multilingual probing of deep pre-trained contextual encoders, *in* Nivre, Joakim, Leon Derczynski, Filip Ginter, Bjørn Lindi, Stephan Oepen, Anders Søgaard, and Jörg Tidemann, editors, *Proceedings of the First NLPL Workshop on Deep Learning for Natural Language Processing*, Linköping University Electronic Press, pp. 37–47. https://aclanthology.org/W19-6205/.

Reimers, Nils and Iryna Gurevych (2019), Sentence-BERT: Sentence embeddings using Siamese BERT-networks, *in* Inui, Kentaro, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Association for Computational Linguistics, Hong Kong, China, pp. 3982–3992. https://aclanthology.org/D19-1410/.

Rogers, Anna, Olga Kovaleva, and Anna Rumshisky (2020), A primer in BERTology: What we know about how BERT works, *Transactions of the Association for Computational Linguistics* **8**, pp. 842–866, MIT Press, Cambridge, MA. https://aclanthology.org/2020.tacl-1.54/.

Sarti, Gabriele, Dominique Brunato, and Felice Dell'Orletta (2021), That looks hard: Characterizing linguistic complexity in humans and language models, *in* Chersoni, Emmanuele, Nora Hollenstein, Cassandra Jacobs, Yohei Oseki, Laurent Prévot, and Enrico Santus, editors, *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*, Association for Computational Linguistics, Online, pp. 48–60. https://aclanthology.org/2021.cmcl-1.5/.

Sennrich, Rico, Barry Haddow, and Alexandra Birch (2016), Neural machine translation of rare words with subword units, *in* Erk, Katrin and Noah A. Smith, editors, *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, Berlin, Germany, pp. 1715–1725. https://aclanthology.org/P16-1162/.

Serikov, Oleg, Vitaly Protasov, Ekaterina Voloshina, Viktoria Knyazkova, and Tatiana Shavrina (2022), Universal and independent: Multilingual probing framework for exhaustive model interpretation and evaluation, *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, Association for Computational Linguistics, pp. 441–456. https://aclanthology.org/2022.blackboxnlp-1.37.

Shi, Xing, Inkit Padhi, and Kevin Knight (2016), Does string-based neural MT learn source syntax?, *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, pp. 1526–1534. http://aclweb.org/anthology/D16-1159.

Shopen, Timothy, editor (2007), *Language Typology and Syntactic Description*, Vol. 1, Cambridge University Press, Cambridge.

Sinnemäki, Kaius (2011), Language universals and linguistic complexity: Three case studies in core argument marking. 978-952-10-7259-8.

Straka, Milan (2018), UDPipe 2.0 prototype at CoNLL 2018 UD shared task, *in* Zeman, Daniel and Jan Hajič, editors, *Proceedings of the CoNLL 2018 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, Association for Computational Linguistics, Brussels, Belgium, pp. 197–207. https://aclanthology.org/K18-2020/.

Strobelt, Hendrik, Sebastian Gehrmann, Hanspeter Pfister, and Alexander M. Rush (2018), Lstmvis: A tool for visual analysis of hidden state dynamics in recurrent neural networks, *IEEE Transactions on Visualization and Computer Graphics* **24** (1), pp. 667–676, IEEE Computer Society.

Tenney, Ian, Dipanjan Das, and Ellie Pavlick (2019a), BERT rediscovers the classical NLP pipeline, *in* Korhonen, Anna, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Florence, Italy, pp. 4593–4601. https://aclanthology.org/P19-1452/.

Tenney, Ian, Patrick Xia, Berlin Chen, Alex Wang, Adam Poliak, R. Thomas McCoy, Najoung Kim, Benjamin Van Durme, Samuel R. Bowman, Dipanjan Das, and Ellie Pavlick (2019b), What do you learn from context? probing for sentence structure in contextualized word representations, *International Conference on Learning Representations*. https://openreview.net/forum?id=SJzSgnRcKX.

Tikhonova, Maria, Vladislav Mikhailov, Dina Pisarevskaya, Valentin Malykh, and Tatiana Shavrina (2023), Ad astra or astray: Exploring linguistic knowledge of multilingual bert through nli task, *Natural Language Engineering* **29** (3), pp. 554–583, Cambridge University Press.

Waldis, Andreas, Vagrant Gautam, Anne Lauscher, Dietrich Klakow, and Iryna Gurevych (2025), Aligned probing: Relating toxic behavior and model internals. https://arxiv.org/abs/2503.13390.

Wangperawong, Artit (2022), Multilingual search with subword TF-IDF. http://arxiv.org/abs/2209.14281.

Zhu, Zining, Soroosh Shahtalebi, and Frank Rudzicz (2022), Predicting fine-tuning performance with probing, *in* Goldberg, Yoav, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, pp. 11534–11547. https://aclanthology.org/2022.emnlp-main.793/.

Şahin, Gözde Gül, Clara Vania, Ilia Kuznetsov, and Iryna Gurevych (2020), LINSPECTOR: Multilingual probing tasks for word representations, **46** (2), pp. 335–385, MIT Press. https://direct.mit.edu/coli/article/46/2/335-385/93365.